Designing Microsoft Azure Infrastructure Solutions: AZ:305

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TECHNOLOGY

Design Migrations

Learning Objectives

By the end of this lesson, you will be able to:

- Evaluate Azure Migrate to enable a simplified approach to migration, modernization, and optimization within Azure
- Identify the stages involved in Microsoft Cloud Adoption Framework for Azure to support users in driving Azure adoption across their organizations
- Use the Cloud Adoption Framework to assess, migrate, optimize, and monitor the Azure migration path
- Analyze Azure Database Migration Service to facilitate both online and offline migrations from various databases

Learning Objectives

By the end of this lesson, you will be able to:

- Examine the workflow of AzCopy for copying blobs or files to or from a storage account
- List the benefits of Azure File Sync to enable the centralization of the organization's file shares

TECHNOLOGY

Azure Migrate

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Azure Migrate

It helps users migrate on-premise VMware VMs, Hyper-V VMs, physical servers, other virtualized machines, and private or public cloud instances to Azure.

Key Features of Azure Migrate

- Assessment and discovery
- Compatibility analysis
- Performance-based sizing
- Migration tools
- Cost estimation
- Integration with Azure resource mover
- Agentless discovery
- Dependency mapping
- Customizable assessments
- Security and compliance

Cloud Migration in the Cloud Adoption Framework

The Microsoft Cloud Adoption Framework for Azure is available to support users in driving Azure adoption across their organizations. The key stages involved are:

- Define strategy
- Plan
- Ready
- Migrate
- Innovate
- Govern
- Manage
- Organize

Cloud Migration in the Cloud Adoption Framework

Define strategy	Define the business case for adoption as well as the projected outcomes
Plan	Align actionable adoption plans with organizational goals
Ready	Prepare the cloud environment for the changes that will be implemented
Migrate	Migrate and modernize existing workloads

Cloud Migration in the Cloud Adoption Framework

Innovate	Create innovative hybrid or cloud-native solutions
Govern	Ensure that the environment and workloads are well-managed
Manage	Supports cloud and hybrid solution operations management
Organize	Establishes teams and responsibilities to support cloud adoption activities

VMware Migration

VMware VMs can be migrated using Azure Migrate Server Migration with the following two options:

Agentless migration

Allows migration of VMs without installing an agent on the source VMs

Agent-based migration

Involves the installing of the Azure Site Recovery Mobility service agent on the source VMware VMs

Agentless Migration: Limitations

Limited application awareness can lead to issues with complex dependencies.

High network bandwidth requirements can impact performance.

Compatibility constraints may restrict usability.

Lack of granular control limits customization.

Agent-Based Migration: Limitations

Agent installation is required on source VMs.

Maintenance overhead includes agent updates.

Compatibility issues may arise with certain systems.

Deployment Tasks Comparison: Agentless vs. Agent-Based

Task	Details	Agentless	Agent-based
Prepare VMware servers and VMs for migration	Configure a number of settings on VMware servers and VMs	Required	Required
Add the server migration tool	Add the Azure Migrate Server Migration tool to the Azure Migrate project	Required	Required
Deploy the Azure Migrate appliance	Set up a lightweight appliance on a VMware VM for VM discovery and assessment	Required	Not required
Install the mobility service on VMs	Install the mobility service on each VM the user wants to replicate	Not required	Required

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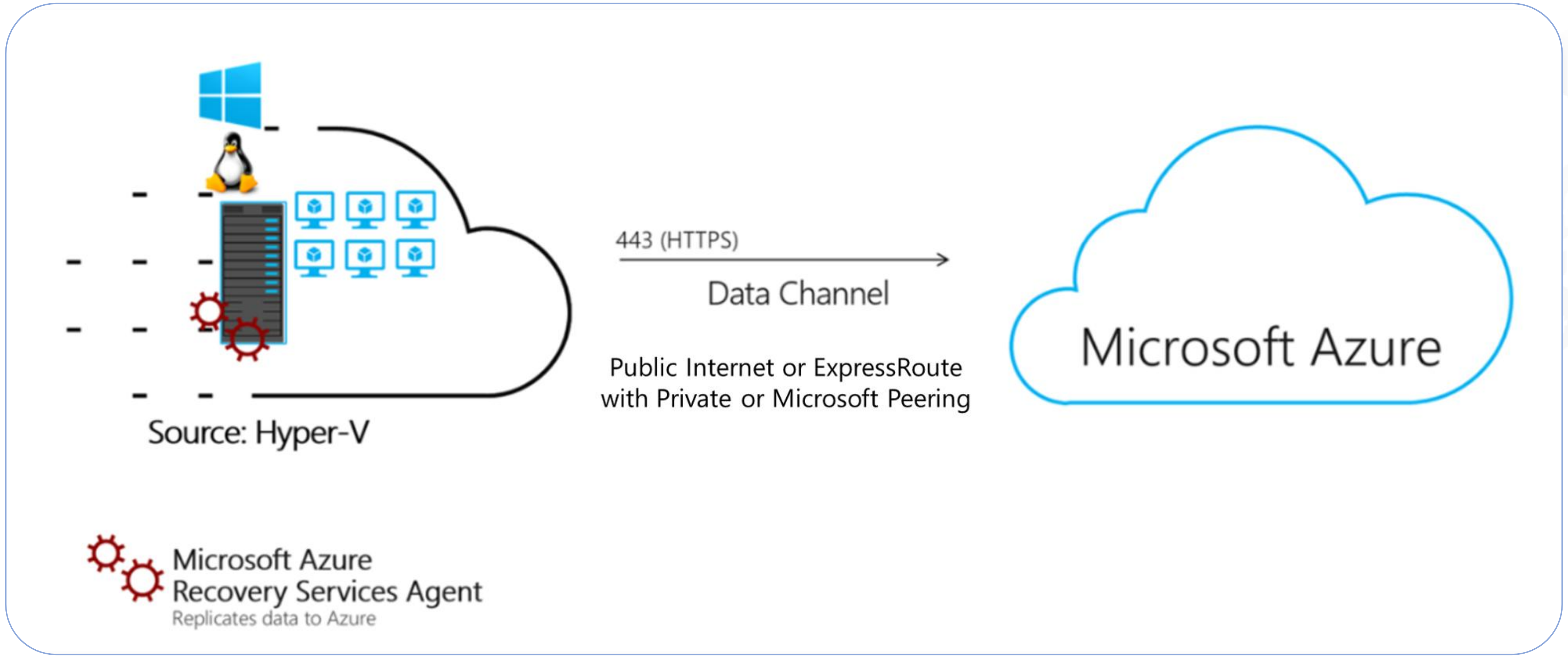
Deployment Tasks Comparison: Agentless vs. Agent-Based

Task	Details	Agentless	Agent-based
Deploy the Azure Migrate Server	Set up an appliance on a VMware VM to discover VMs, and bridge between the mobility service running on VMs and Azure Migrate Server Migration	Not required	Required
Enable VM replication	Configure replication settings and select VMs to replicate	Required	Required
Run a test migration	Run a test migration to make sure everything is working as expected	Required	Required
Run a full migration	Migrate the VMs	Required	Required

Agentless Migration: Architectural Components

It provides agentless replication for on-premises Hyper-V VMs, using an optimized migration workflow for Hyper-V.



Components: Replication provider and Recovery Services agent

Source: https://learn.microsoft.com/en-us/azure/migrate/hyper-v-migration-architecture

Agentless Migration: Architectural Components

Replication provider

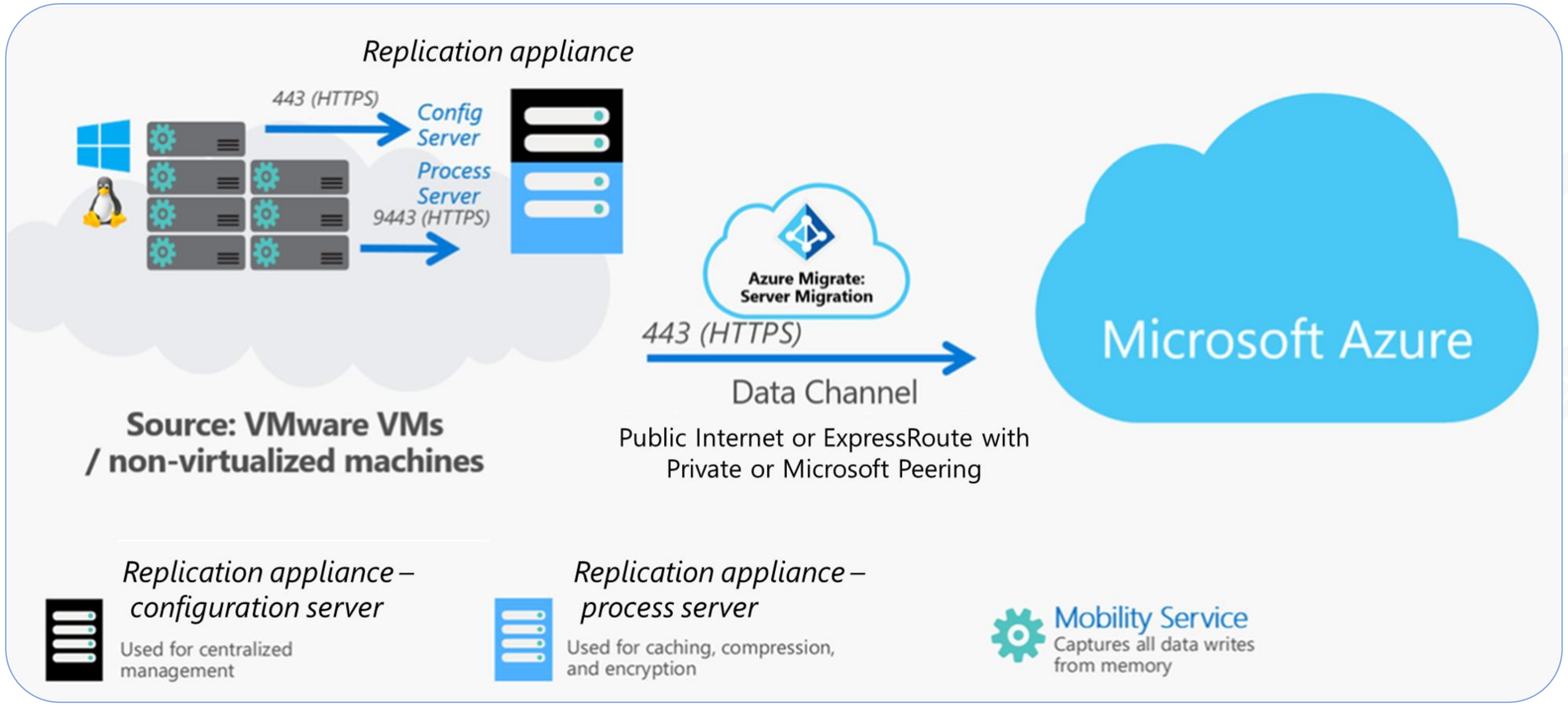
It is installed on Hyper-V hosts and registered with Azure Migration Server Migration, orchestrating replication for Hyper-V VMs.

Recovery Services agent

It handles data replication by working with the provider to replicate data from Hyper-V VMs to Azure.

Agent-Based Migration: Architectural Components

It is used to migrate on-premises VMware VMs and physical servers to Azure.



Components: Replication appliance and Mobility service

Source: https://learn.microsoft.com/en-us/azure/migrate/agent-based-migration-architecture

Agent-Based Migration: Architectural Components

Replication appliance

It is an on-premises server that acts as a bridge between the on-premises environment, and the Migration and modernization tool.

Mobility service

It is an agent installed on each server you want to replicate and migrate. It sends replication data from the server to the process server.

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Assessments Using Azure Migrate

Planning Azure Migration

A user can follow the migration path by using the Cloud Adoption Framework to:

Assess
Start with a full assessment of the current environment.

Optimize
Once the services are migrated, it is important to optimize them to ensure their efficiency.

Migrate
Destination systems and services on Azure are required for migration.

Monitor
The Log Analytics agent captures health and performance information from Azure VMs.

Assess

Commence with a thorough assessment of the current environment, identifying servers, applications, and services within the migration scope

Assess

For each application, there are multiple migration options, such as:

Rehost	Recreate existing infrastructure by moving virtual machines from the data center to virtual machines on Azure
Refactor	Move services running on VMs to PaaS services
Rearchitect	Rearchitect some systems so that they can be migrated
Rebuild	Rebuild software if the cost to rearchitect is more than that of starting from scratch
Replace	Third-party applications could replace custom applications. Evaluate SaaS options to replace existing applications

Assess

Migration planning includes the following steps:

01 Create a cloud migration plan (Requirements, environment, and tools)

02 Involve stakeholders (Business and IT)

03 Estimate cost savings using tools like TCO (Total Cost of Ownership)

04 Discover, evaluate, and document applications

05 Discover tools that help in the migration journey

Migrate

Migration includes the following steps:

Step 01
Prepare the source and target environments

Step 02
Set up and start replication

Step 03
Test replication

Step 04
Migrate to Azure

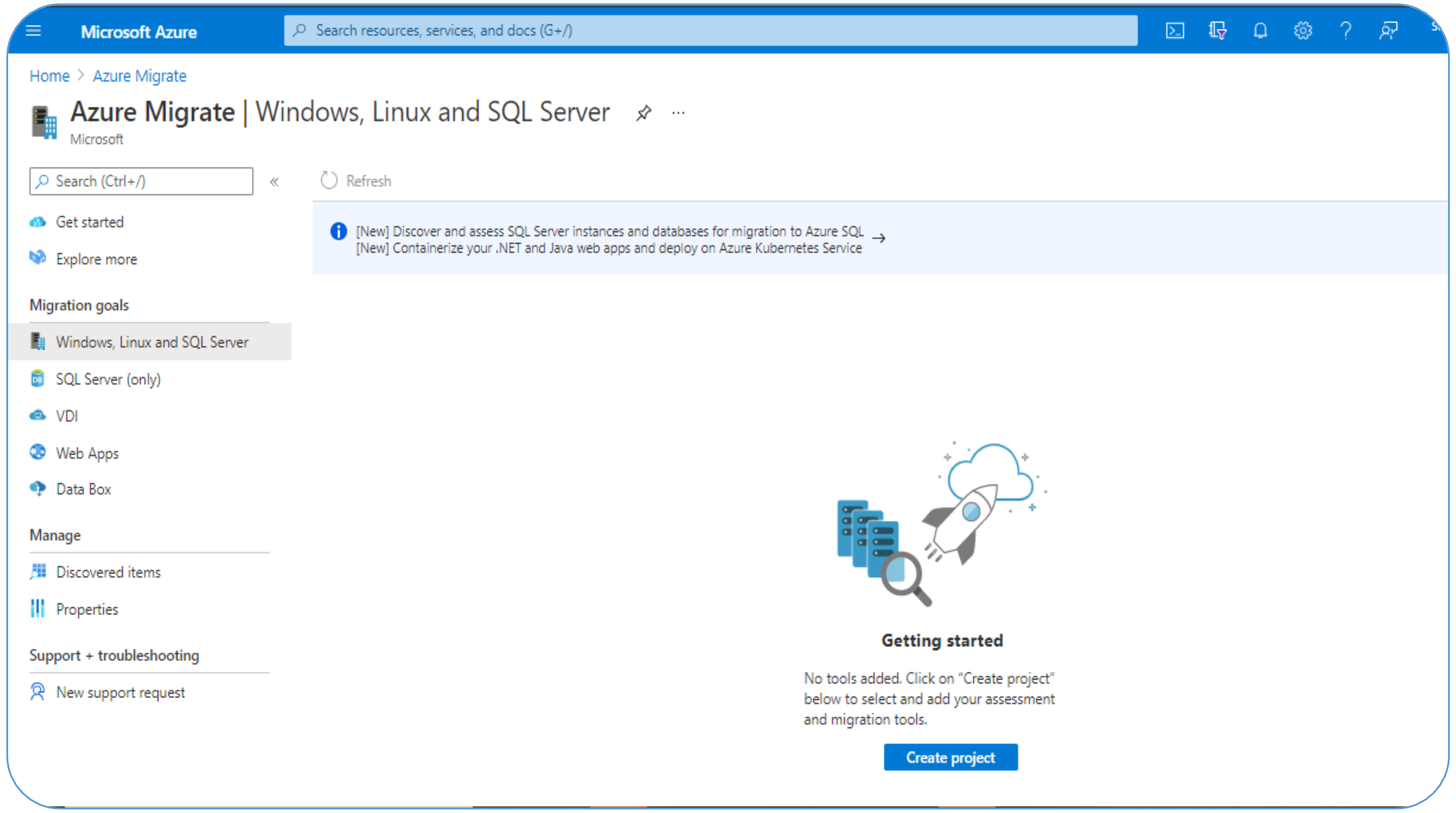
Migrate

Database migration includes the following steps:

- Assess the on-premises databases
- Migrate the schemas
- Create and run an Azure Database Migration Service
- Monitor the migration

Using Azure Migrate to Assess Environment

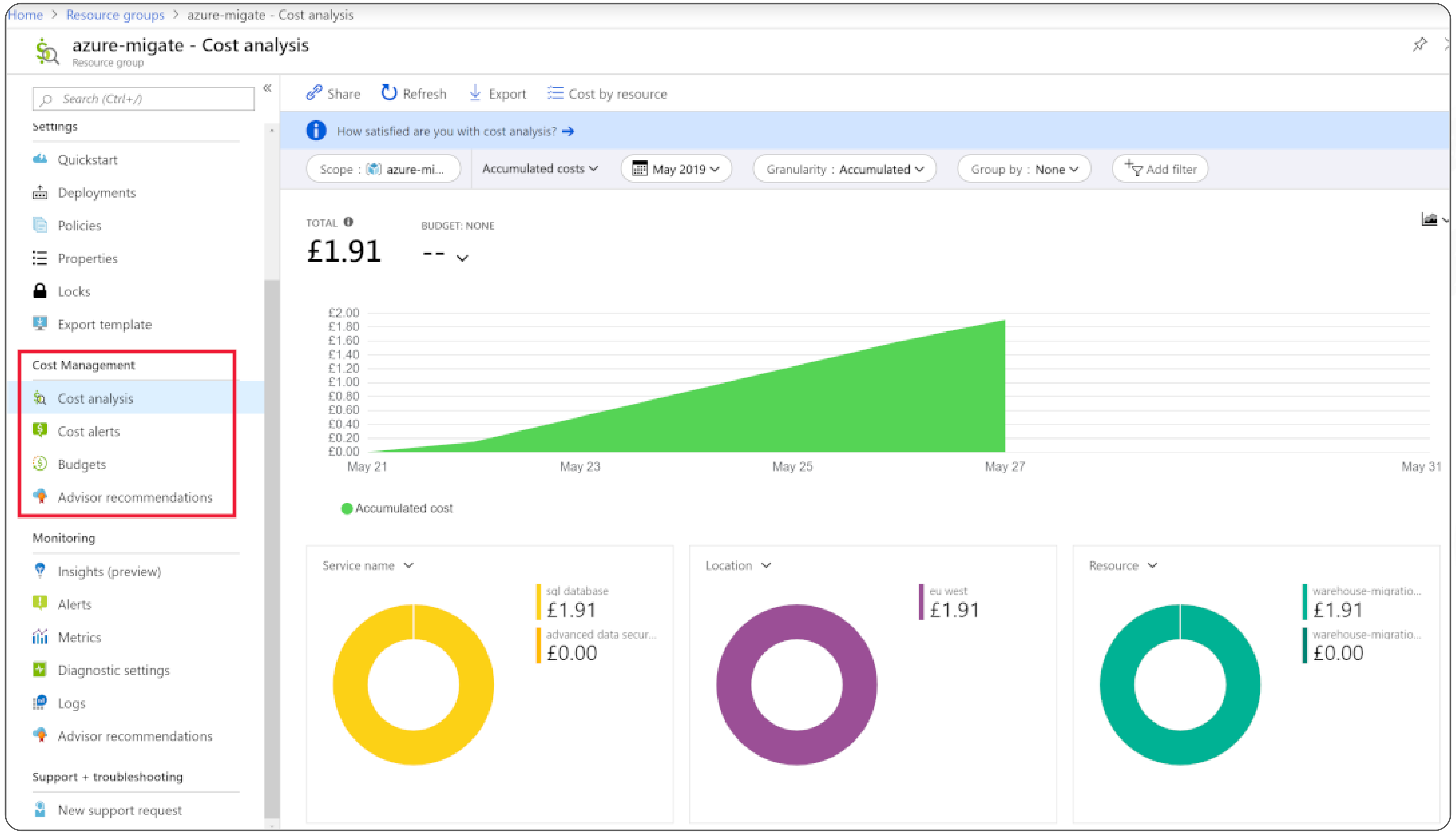
The Azure Migrate assessments are created within a project.



- Group VMs according to the types of VM workloads
- Assessment steps:
  - Discover virtual machines
  - Create assessments based on customization

Optimize

Use Azure Cost Management to analyze Azure costs for various management scopes.



Monitor

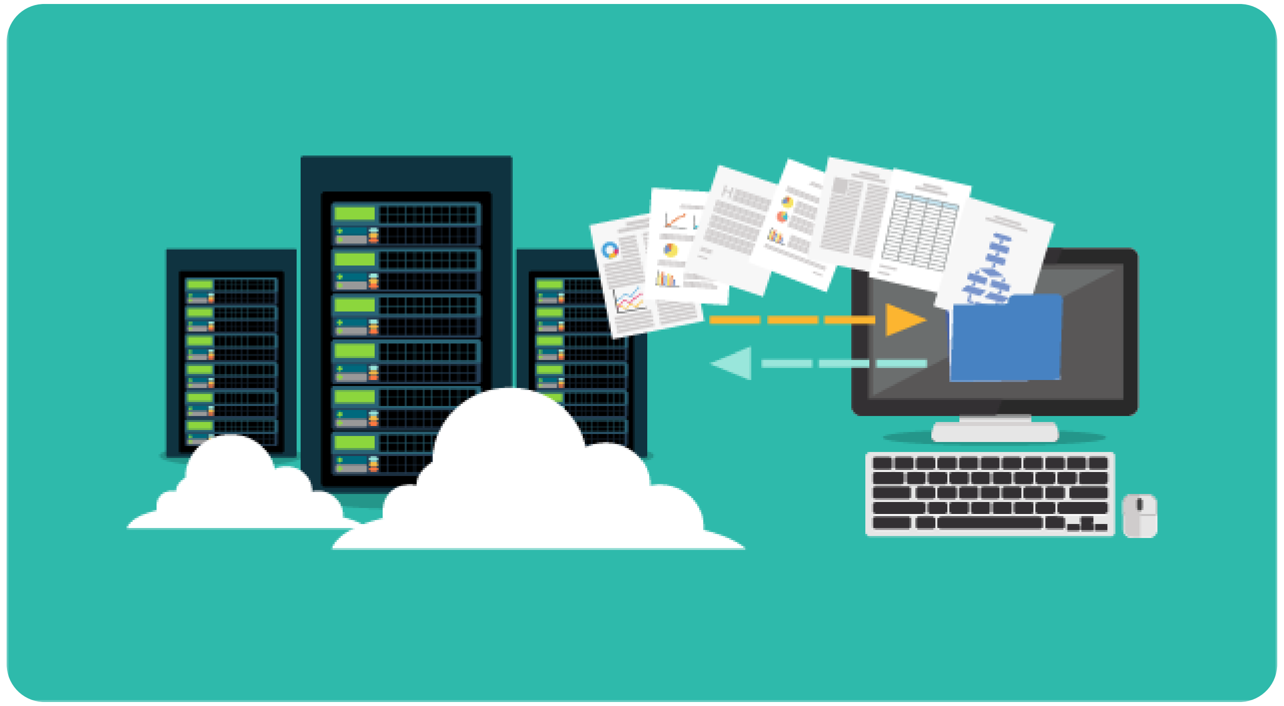
The Azure Monitor captures health and performance information from Azure VMs (Log Analytics agent).

It establishes alerts based on various data sources, such as:

- CPU usage metrics
- Specific text in log files
- Health metrics
- Autoscale metrics

Azure Migrate Service

It assesses readiness and assists with migration to Azure from an on-premises environment.



It helps with performance-based sizing calculations for the machines that will be migrated and estimation of the ongoing cost of running these machines on Azure.

Discover Machines

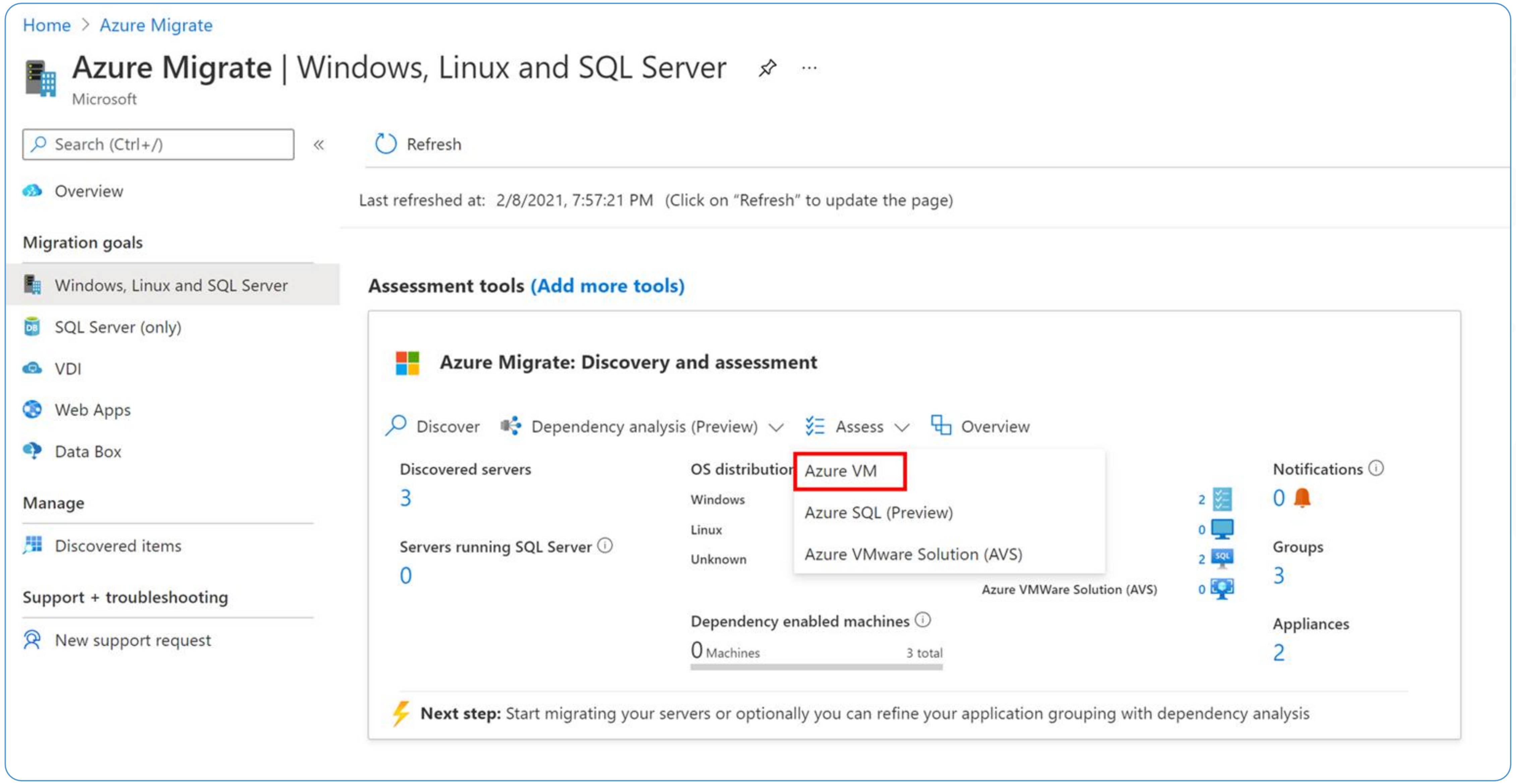
Import and spin up the collector appliance on a Hyper-V or VMware host

- Collector gathers data on VM cores, memory, disk sizes, and network adapters.
- Collect performance data, including CPU and memory usage, disk IOPS, disk throughput, and network.
- Push the collected data to the Azure Migrate project.

The user can view discovered systems on the Azure portal.

Create an Assessment

Azure Migrate assesses an environment's readiness to be migrated to Azure on the portal.



Recommend a Solution for Migrating Workloads to IaaS and PaaS

Migrate Azure App Service with Azure Migration Assistant

Enter a public URL and access the web app for migration:

Get a free compatibility report for your external app

Run a scan on your web app's public URL for a report of the technologies it uses and whether App Service fully supports them. If compatible, you'll be guided to download the migration assistant to simplify your migration.

Assess your web app for migration now.

Enter a public URL (https://)

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Assess

Assess and migrate your on-premises .NET, Java, and Linux web apps to Azure

Download the App Service migration assistant—a fast, free, and automated way to migrate web apps with minimal or no code changes. Run readiness checks and get potential remediation steps for common issues. Receive step-by-step guidance for moving your web app to App Service.

- Simplify the migration of web apps to the cloud with minimal or no code changes
- Run readiness checks and get potential remediation steps for common issues

Migrate Servers with Azure Migrate

The Azure Migrate assessment identifies candidates for server migration to Azure.

Azure Migrate: Server Assessment

Discover Assess Overview

Quick start

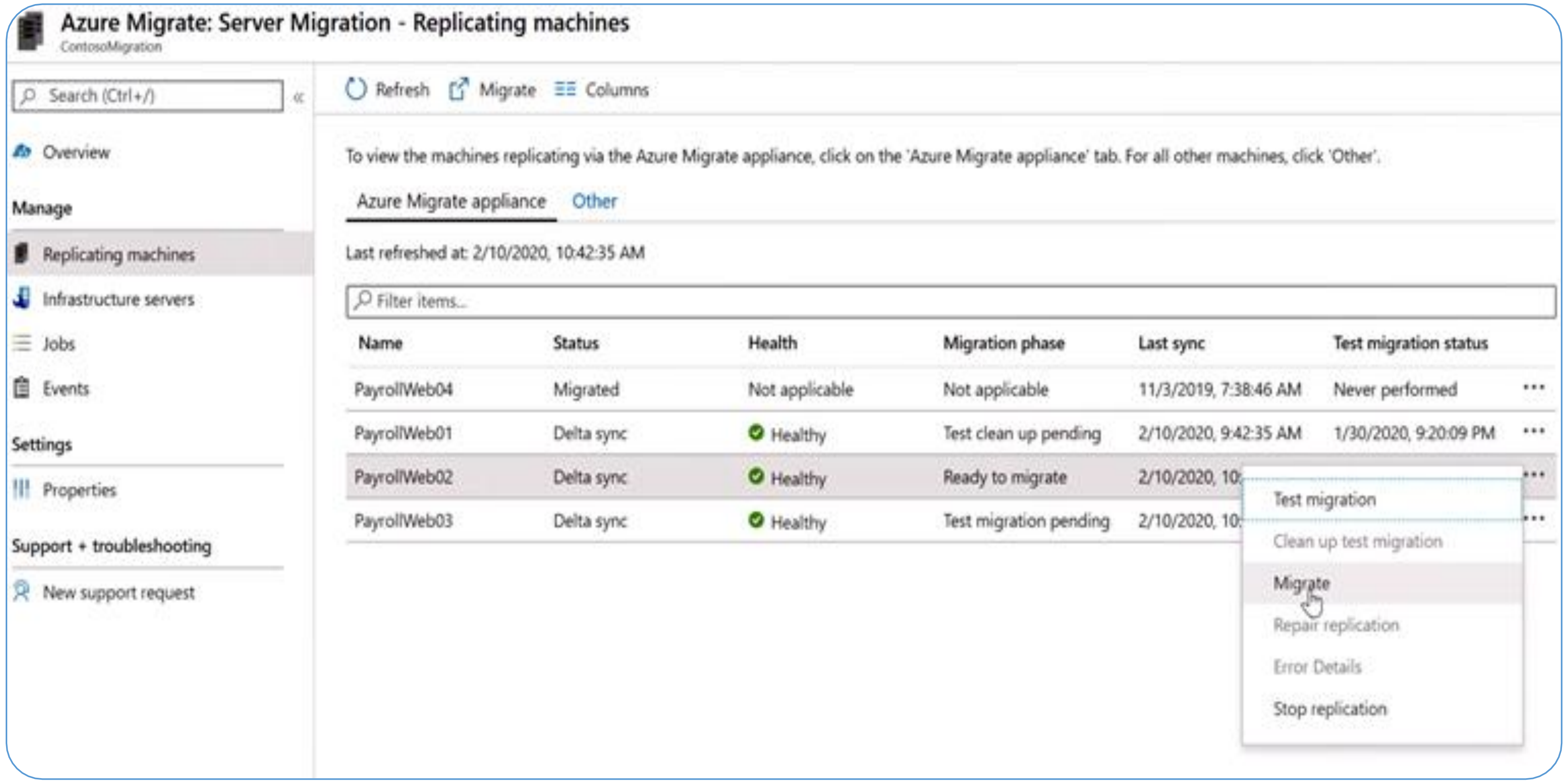
1: Discover
Discover your on-premises machines by using an appliance or importing in a CSV format. Click ‘Discover’ to get started.

2: Assess
Click ‘Assess’ to assess the discovered machines.

Azure Migrate runs an agentless migration of virtual and physical servers into Azure.

Source: https://cloudoasis.com.au/2020/03/17/migrate-your-workload-to-the-cloud-using-azure-migrate/

Migrating the Virtual Machines to Production



Once ready for production migration:

- Select Migrate from the replicating machines
- Shut down the VMs for final replication
- Migrate during off-peak hours

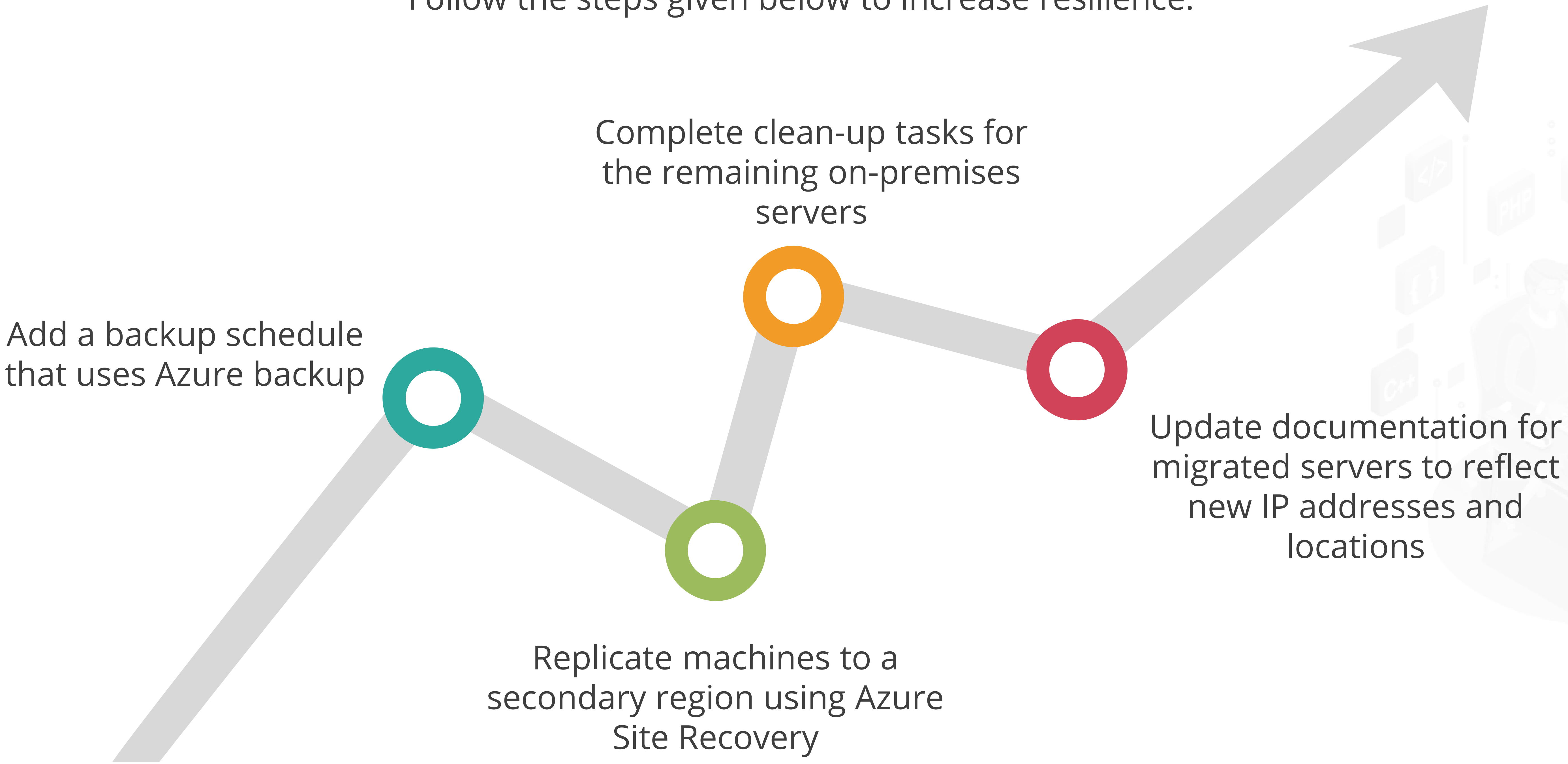
Post-Migration Steps

After the migration:

- Review security settings of virtual machines
- Restrict network access for unused services (NSG)
- Deploy Azure Disk Encryption

Post-Migration Steps

Follow the steps given below to increase resilience:



Migrating Azure App Service with Azure Migration Assistant

Duration: 10 Min.

ASSISTED PRACTICE

Problem Statement:

As an Azure Architect, you have been asked to assist your company with an Azure migration solution for .NET and PHP web projects.

Outcome:

You will be able to successfully demonstrate the use of Azure Migration Assistant to streamline the migration process for .NET and PHP web projects to Azure, ensuring minimal downtime and optimized resource allocation.

Note: Refer to the demo document for detailed steps:
01_Migrating_Azure_App_Service_with_Azure_Migration_Assistant

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create a web app
2. Get a report for the external web app

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Recommend a Solution for Migrating Databases

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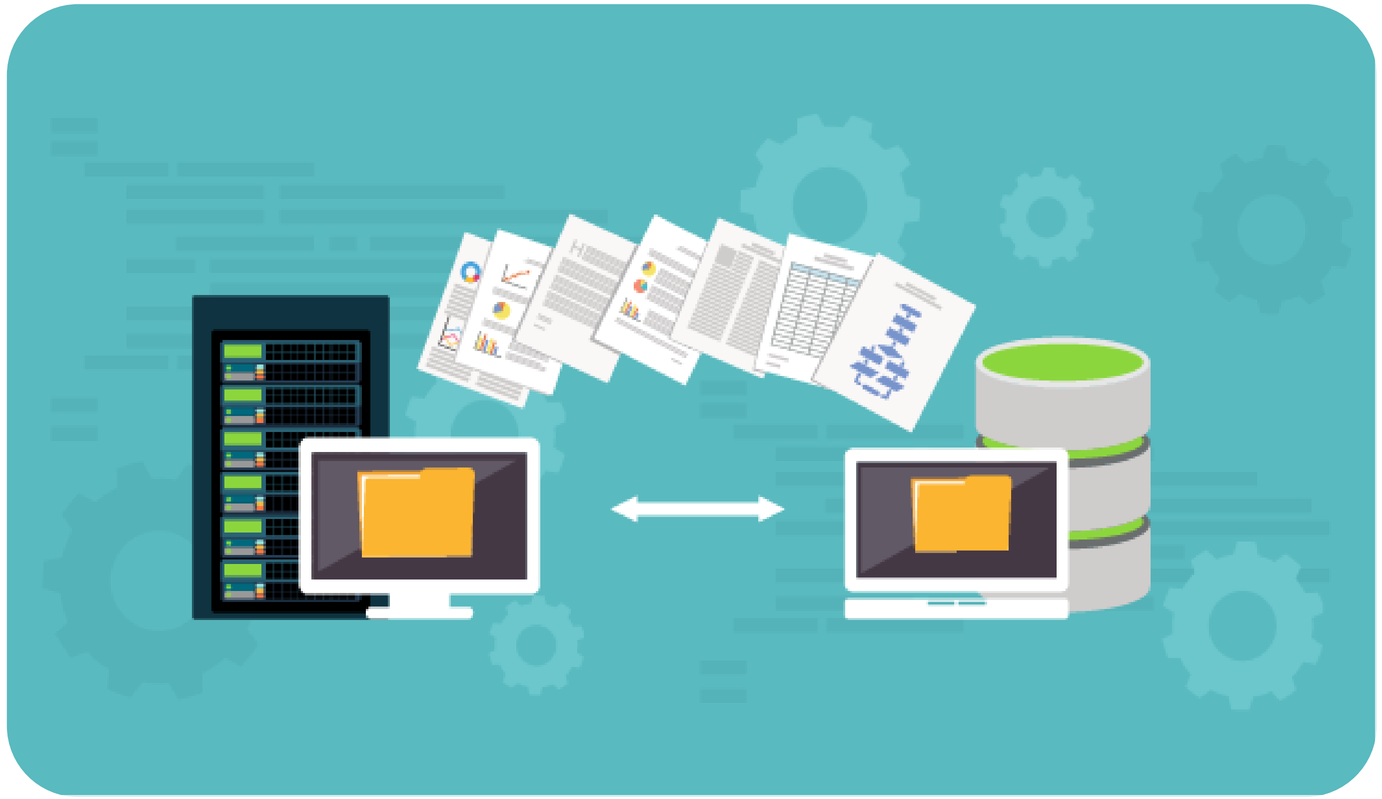
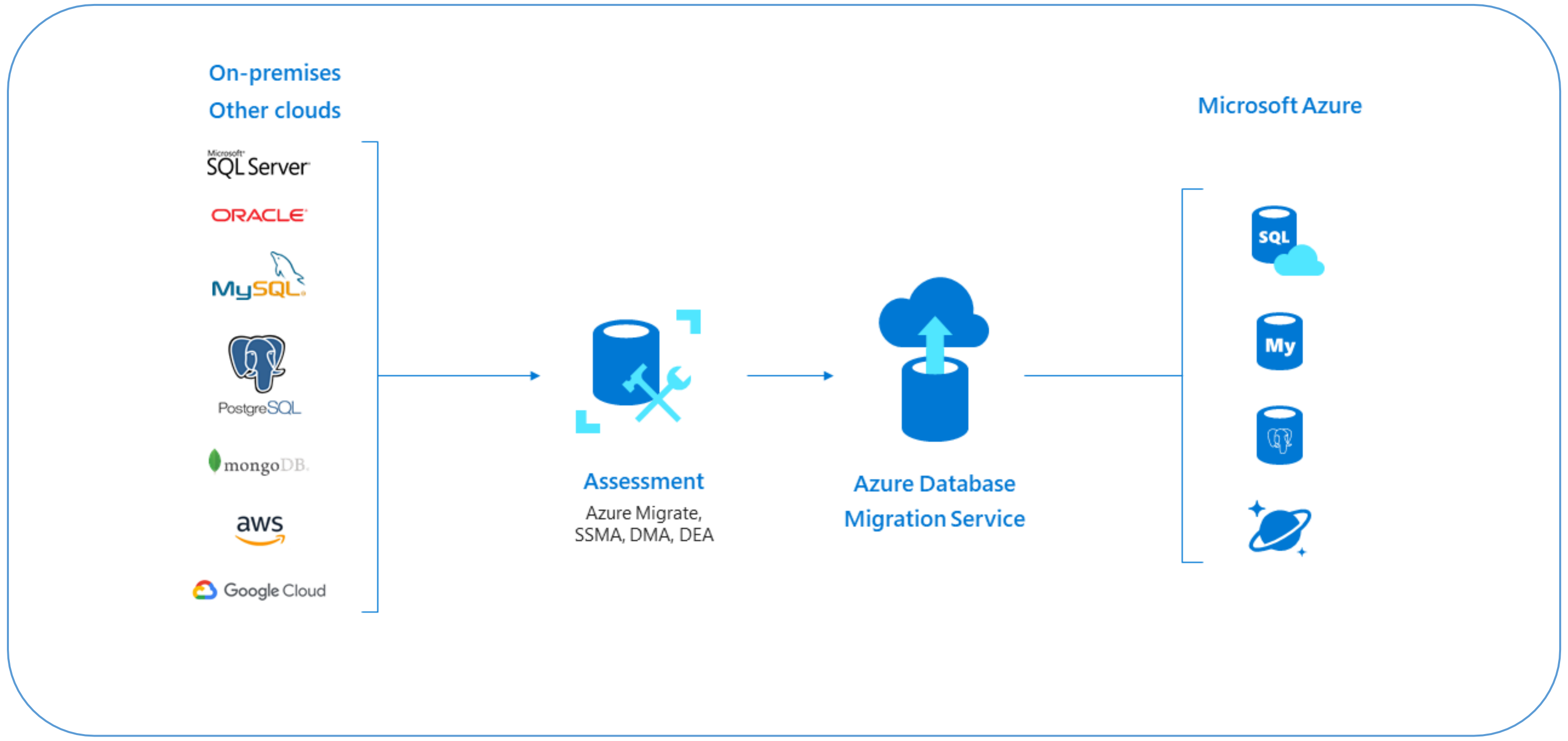
Azure Database Migration Service

It enables online and offline migrations from multiple database sources to Azure data platforms, all with minimal downtime.



Migrate Databases with Azure Database Migration Service

Two different ways to migrate SQL Server databases:



- Offline migration requires shutting down the server at the start of the migration for service.
- Online migration uses a continuous synchronization of live data, allowing a cutover to the Azure replica database at any time.

Source: https://azuremarketplace.microsoft.com/es-es/marketplace/consulting-services/webcom.az001

Migrate Databases with Azure Database Migration Service

User relational database can be migrated to several destinations in Azure, such as:

Single Azure SQL database instance:	A fully managed, single SQL database
Azure SQL database managed instance:	Database Engine but missing some minor SQL Server features
SQL server on Azure virtual machines:	An infrastructure-as-a-service (IaaS) offering that runs a full version of SQL Server and supports all the features of SQL Server

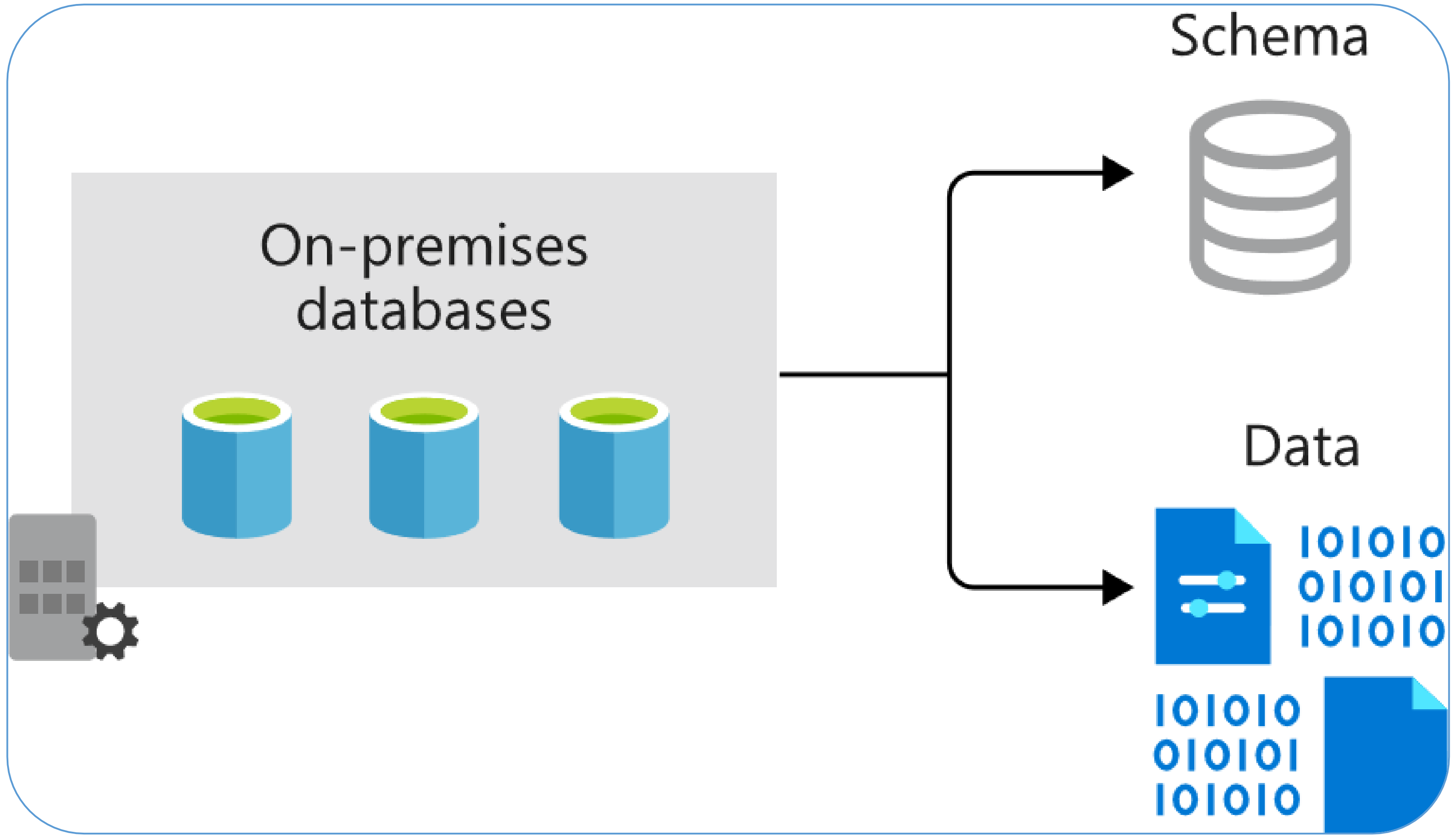
Migrate Databases with Azure Database Migration Service

User relational database can be migrated to several destinations in Azure, such as:

Azure database for MySQL:	An Azure database service based on the MySQL Community Edition, versions 5.6 and 5.7
Azure Database for PostgresSQL:	An Azure database service based on the community version of the PostgreSQL database engine
Azure cosmos DB:	A globally distributed, multi-model, fully managed database service

Overview of Database Migration

Offline and online migrations prerequisites:



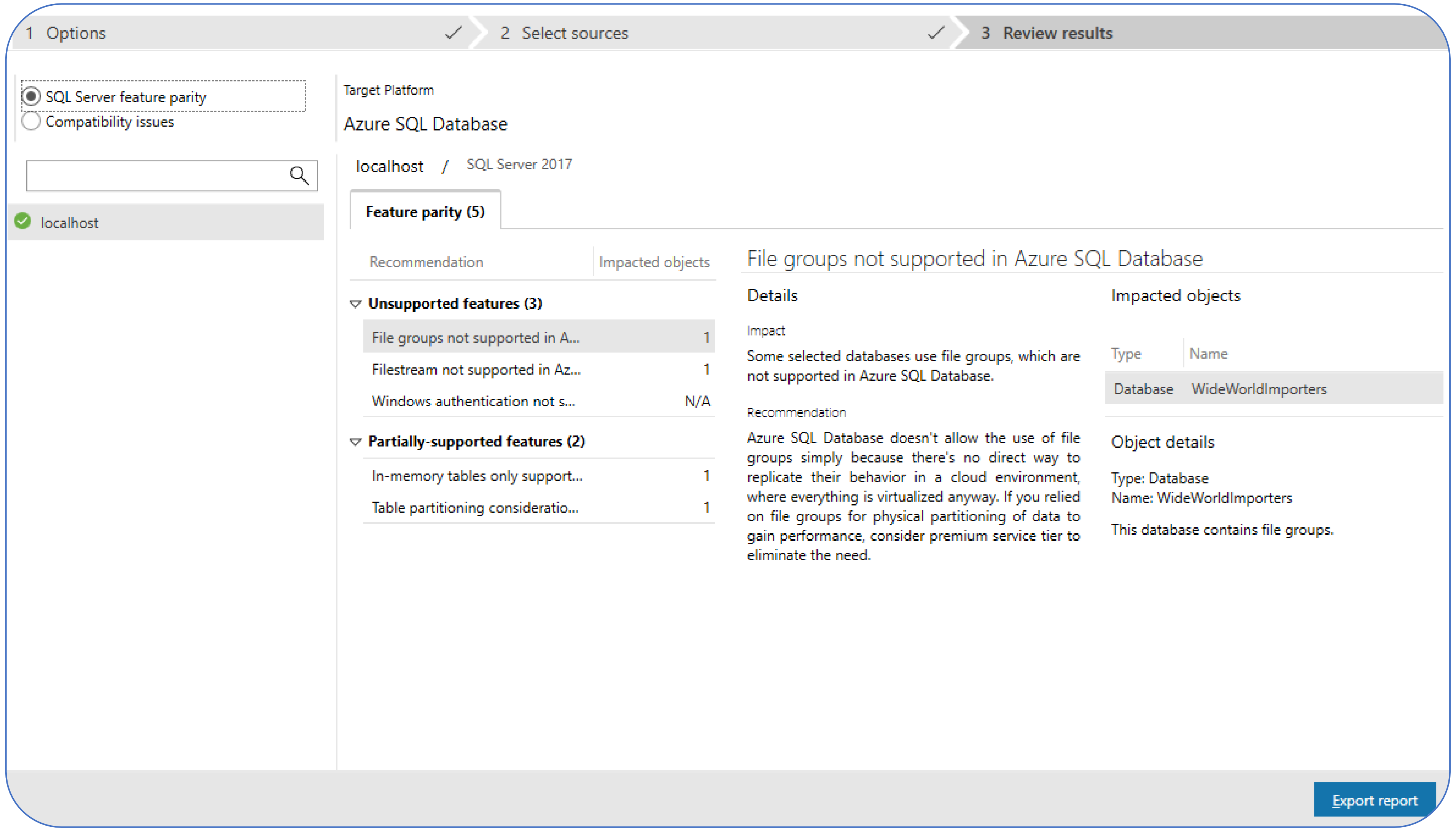
Overview of Database Migration

Offline and online migrations prerequisites:

- Download the Data Migration Assistant
- Create an Azure Virtual Network instance
- Configure the network security group
- Configure the Windows Firewall
- Configure credentials

Assess the On-Premises Databases

Steps to assess the on-premises databases are:



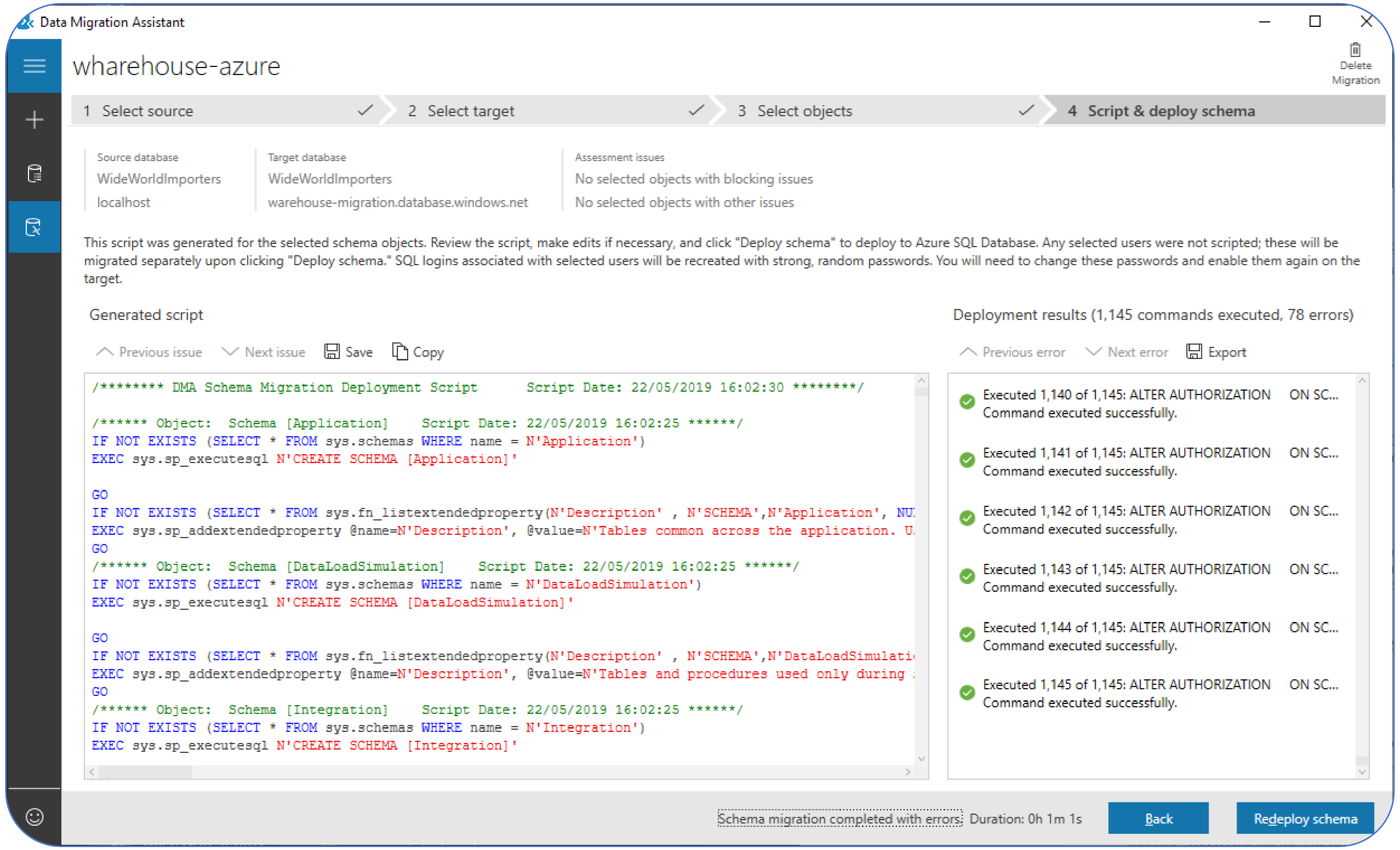
Assess the On-Premises Databases

Steps to assess the on-premises databases are:

- Use Data Migration Assistant to create an Assessment project
- Select the source and target servers
- Provide the connection details and permissions
- Choose the database to migrate

Data Migration Assistant

Steps to migrate the schema using the data migration assistant are:



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Data Migration Assistant

Steps to migrate the schema using the data migration assistant are:

- Select an on-premises SQL Server instance as the source server
- Set the Azure SQL Database instance as the target server
- Set the scope of the migration to Schema Only
- Once connected to the source database, choose the schema objects to deploy to the new SQL database

Steps: Migrate Data with Database Migration Service

01	Create an instance of Azure Database Migration Service
02	Create a new migration project
03	Specify source and target server details
04	Identify the databases
05	Run and monitor the migration
06	Review migrated content

Data Migration: Best Practices

Perform a thorough assessment	Conduct an assessment to identify compatibility issues, unsupported features, and potential blockers
Plan migration in phases	Break the migration into phases (schema, data, application) to manage complexity and risks
Optimize source database	Clean up unnecessary data, indexes, and resolve performance issues before migrating
Choose the right strategy	Select **offline** for quick migrations or **online** for minimal downtime during critical migrations.

Data Migration: Best Practices

Backup and restore data	Backup the source database before migration and test the recovery process
Use Azure DMS	Leverage Azure Database Migration Service (DMS) to automate larger database migrations
Test schema migration	Migrate and validate the schema on the target database first to ensure consistency
Verify data integrity	Run validation scripts post-migration to ensure the accuracy and completeness of data

Data Migration: Best Practices

Monitor migration progress	Use monitoring tools to detect and resolve bottlenecks or errors early in the process
Plan for downtime	Communicate downtime requirements and schedule migrations during low-traffic periods
Document the migration	Keep logs and document migration steps, issues faced, and resolutions for future reference
Post-migration testing	Perform post-migration performance, data accuracy, and application functionality tests

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Determine Migration Scope

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Supported Scopes of Migration

There are four ways to complete the migration of compute, network, and storage resources, such as:

- Migration of virtual machines (Not in a virtual network)
- Migration of virtual machines (In a virtual network)
- Migration of unattached resources
- Migration of storage accounts

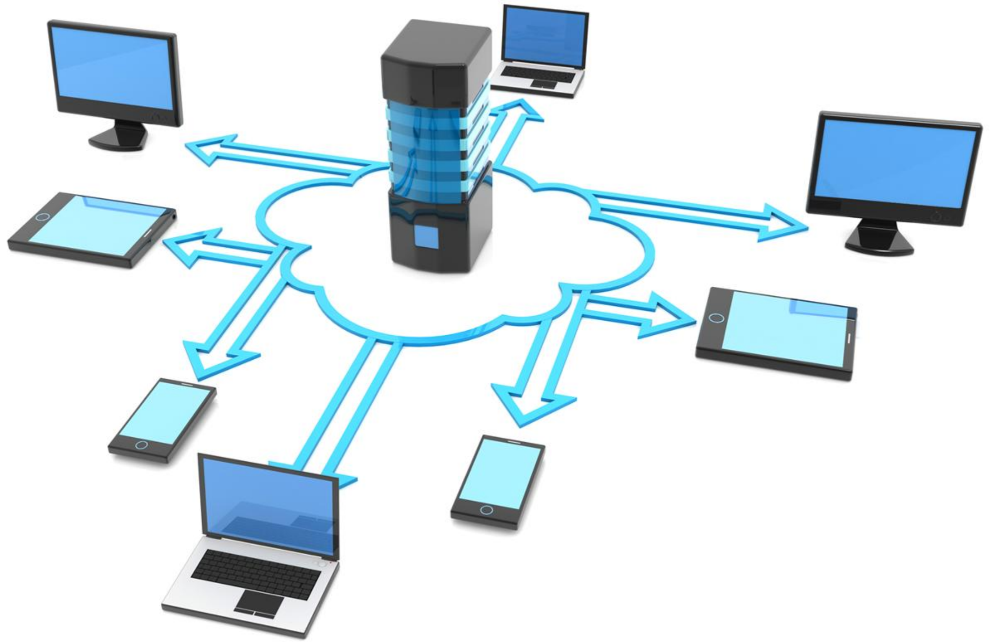
Migration of Virtual Machines (Not in a Virtual Network)

The two options available are:

- Request the platform to create a new virtual network and migrate the virtual machine into the new virtual network
- Migrate the virtual machine into an existing virtual network in Resource Manager

Migration of Virtual Machines (In a Virtual Network)

The two options available are:

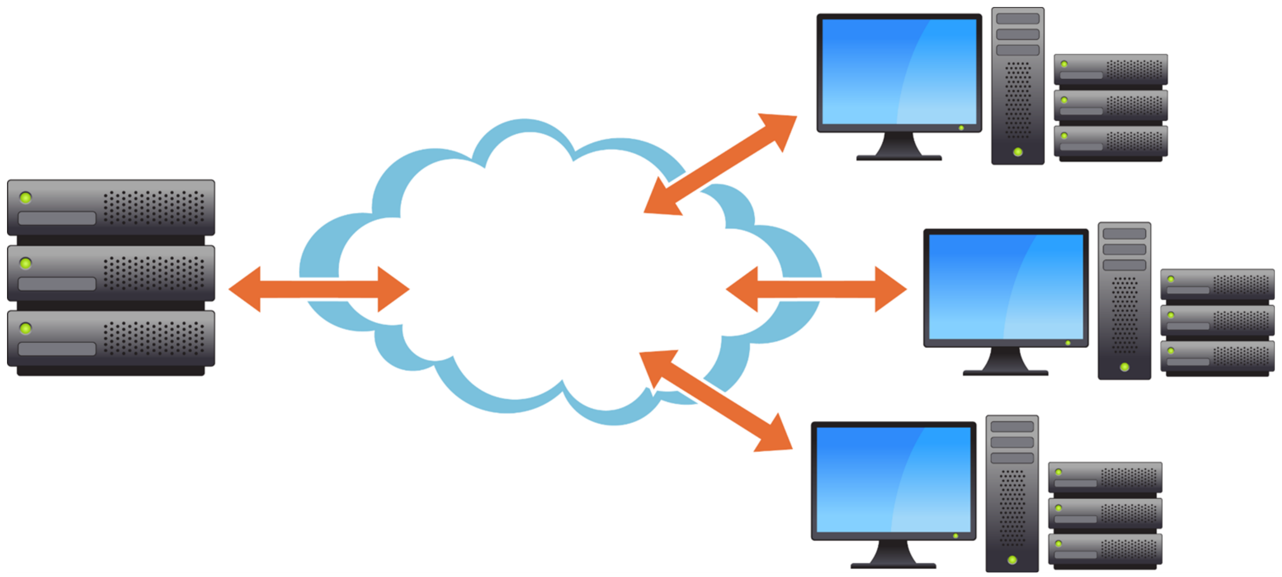


- For most VM configurations, only the metadata migrates between the Classic and Resource Manager deployment models.

- The underlying VMs run on the same hardware, in the same network, and with the same storage.

Migration of Storage Accounts

If users are using a VM with a classic storage account, then compute and network resources should be migrated independently of storage accounts.



To complete the migration procedure, users' storage accounts must be migrated once the virtual machines and virtual network have been migrated.

Migration of Storage Accounts

If a storage account only contains blobs, files, tables, and queues, the migration to Azure Resource Manager can be done independently.

Migration of Unattached Resources

Storage accounts with no associated disks or virtual machine data can be migrated independently.

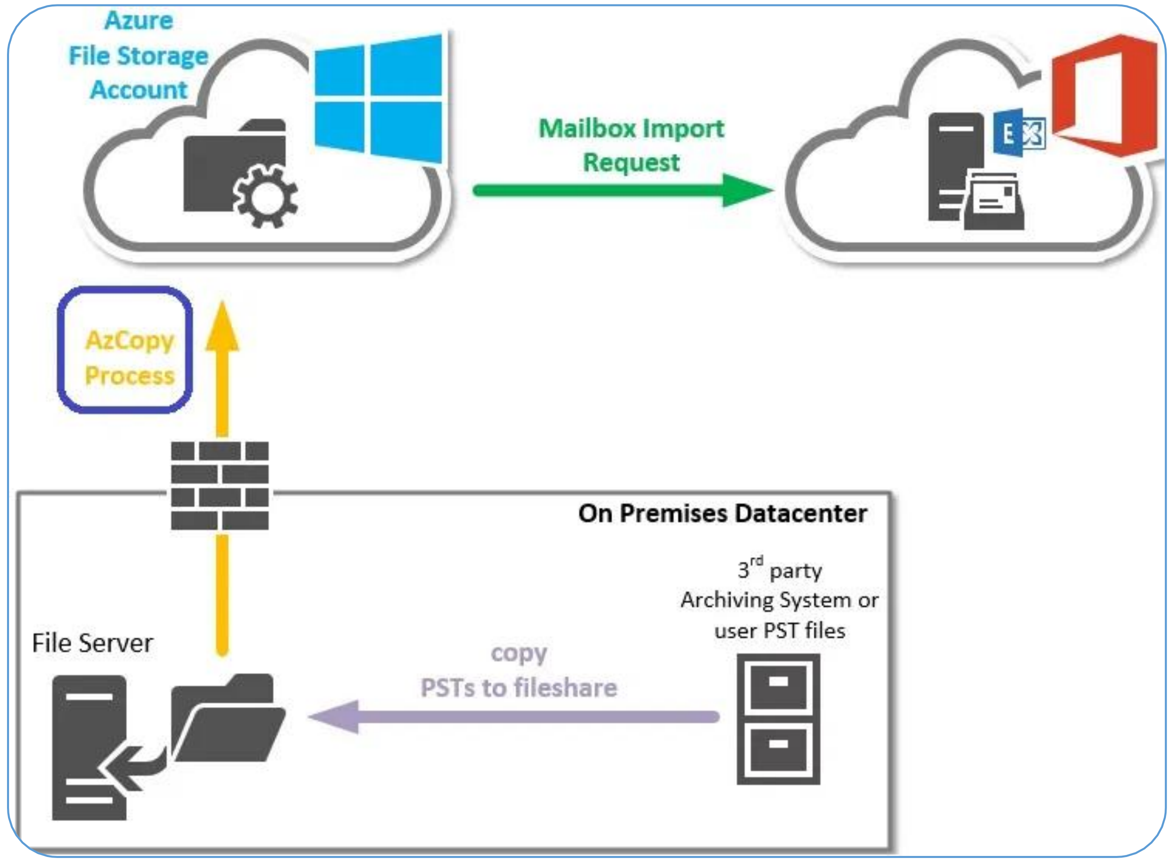
Network security groups, route tables, and reserved IPs are not attached to any virtual machines.

TECHNOLOGY

Recommend a Solution for Migrating Data

AzCopy

It is a command-line tool for copying blobs or files to or from a storage account.



Users can either copy data between a file system and a storage account or between storage accounts using AzCopy.

Source: https://www.systranbox.com/how-to-install-azcopy-on-linux/

Workflow of Using AzCopy

Create a storage account

Use AzCopy to upload all the data

Modify the data for test purposes

Create a scheduled task or cron job to identify new files to upload

Authenticate with SAS Token Using AzCopy

Efficiently copy and authenticate your local files to Azure Blob Storage using AzCopy with a SAS token, ensuring secure and streamlined data transfers.

Syntax:

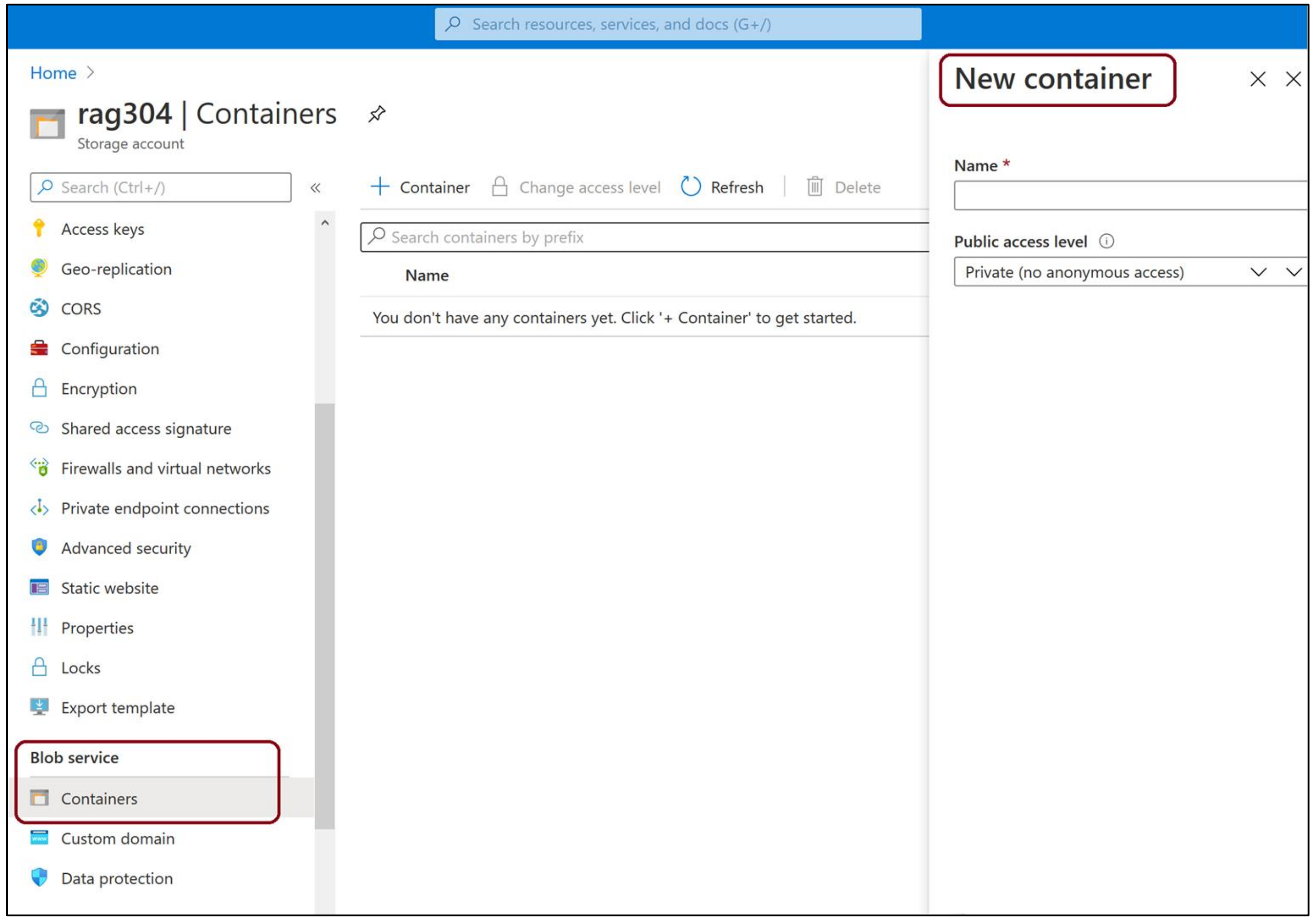
```
azcopy copy "C:/path/to/local/folder" "https://<storage-account-
name>.blob.core.windows.net/<container-name>/<directory-name>?<SAS-
Token>" --recursive
```

Steps: Migrate On-premises Data to Cloud Storage

01	Create a container
02	Authenticate with Entra ID
03	Upload contents of a folder to Blob Storage
04	Upload modified files to Blob Storage
05	Create a scheduled task

Create a Container

Blobs must be uploaded into a container.

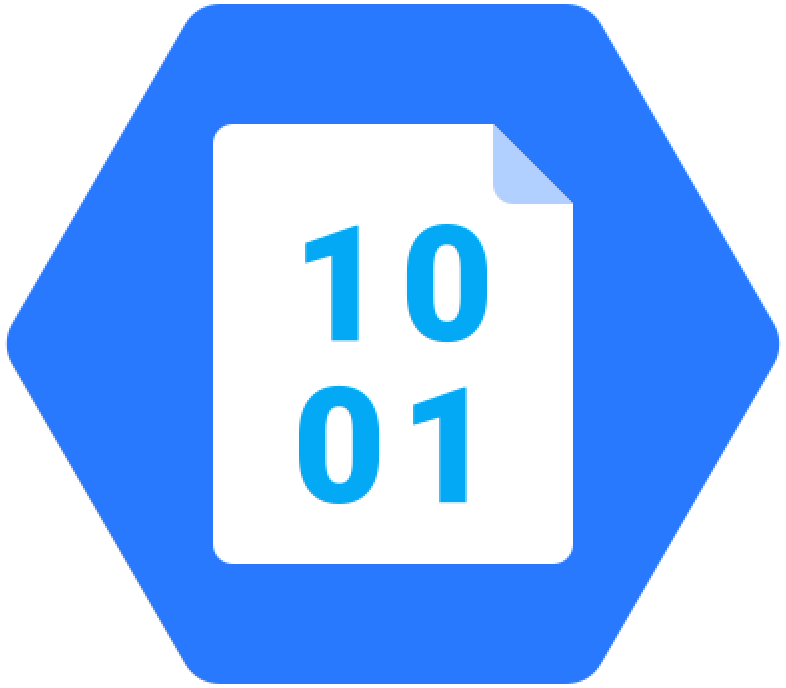


Steps to upload Blobs on a container:

- Select the storage account
- Select Containers under Blob service

Authenticate with Entra ID

Provide access using RBAC (role-based access control) to perform the required operation:



1. Storage Blob Data Contributor
2. AzCopy
3. Authentication URL (code provided)
4. Sign into your Azure account
5. Begin using AzCopy

Upload Contents of a Folder to Blob Storage

The syntax for uploading a file content to Blob storage using the **azcopy copy** command is:

Upload Blob:

```
azcopy copy “<local-folder-
path>" "https://<storage-account-
name>.<blob or dfs>.core.windows.net/<containe
r-name>" --recursive=true
```

To upload the contents of the specified directory to Blob storage recursively, specify the recursive option.

Upload Modified Files to Blob Storage

To upload files based on their last modified time, use the following **AzCopy sync** command:

Syntax

```
azcopy sync "<local-folder-path>"
"https://<storage-account-
name>.blob.core.windows.net/<container-name>"
--recursive=true
```

Create a Scheduled Task

Steps to be followed for creating a scheduled task using AzCopy command script:

01 Copy the AzCopy command to a text editor (Windows or Linux)

02 Update the parameter values

03 Save the file as script.sh (Linux) or script.bat (Windows)

Create a Scheduled Task

To create a cron job using the AzCopy command script:

For Linux, using Terminal:

```
crontab -e */5 * * * * sh
/path/to/script.sh
```

For Windows, using PowerShell:

```
schtasks /CREATE /SC minute /MO 5
/TN "AzCopy Script" /TR
C:\script.bat
```

Migrate Offline Data

Two ways of migrating offline data to Azure:

- Azure Import/Export
- Azure Data Box

Azure Import/Export

It is an Azure service that allows users to move huge amounts of data between on-premises and Azure Storage accounts.

Azure Import/Export Use Cases

Data migration to the cloud	Move large amounts of data to Azure quickly and cost-effectively
Content distribution	Quickly send data to the customer sites
Backup	Take backups of on-premises data to store in Azure Storage
Data recovery	Recover large amounts of data stored in storage and have it delivered to the on-premises location

Azure Data Box

The Azure Data Box cloud solution from Microsoft facilitates the fast, cost-effective, and reliable transfer of terabytes of data to and from Azure.

It can be divided into two groups:

Offline data transfer

Online data transfer

Offline Data Transfer

The devices in the offline grouping include:

Data Box Disk	Data Box	Data Box Heavy
Provides one ~35-TB transfer to Azure	Provides one ~80-TB transfer to Azure per order	Provides one ~800-TB transfer to Azure

Online Data Transfer

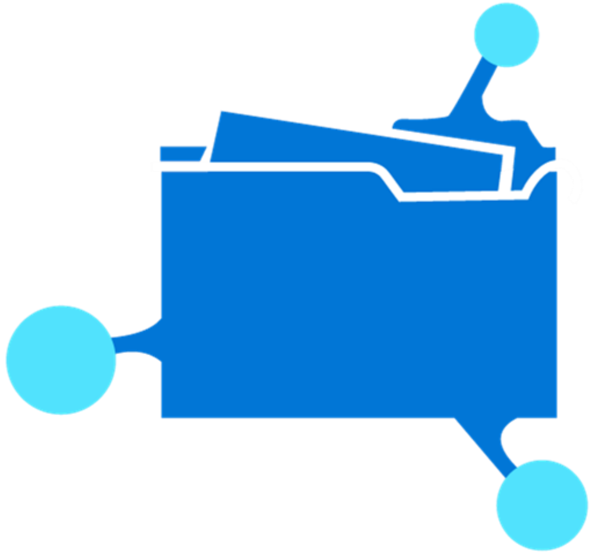
Online data transfer enables a link between a user's on-premises assets and Azure.

Data Box Gateway:

- This device is an entirely virtual appliance.
- It is based on a virtual machine that one can provision in their on-premises environment.

Azure File Sync

It enables centralizing the organization's file shares in Azure Files while keeping the flexibility, performance, and compatibility of a Windows file server.



Source: : https://docs.microsoft.com/

Benefits of Azure File Sync

Cloud tiering:
The most frequently accessed files are cached on the local server, and the least frequently accessed files are tiered to the cloud.

Multi-site access and sync:
It is good for distributed access scenarios.

Business continuity and disaster recovery:
Azure Files backs the Azure file sync, which offers several redundancy options for highly available storage.

Key Takeaways

- Azure Migrate service assesses readiness and assists with migration to Azure from an on-premises environment.
- Azure Migrate runs an agentless migration of virtual and physical servers into Azure.
- Azure DB migration service enables online and offline migrations from multiple database sources to Azure data platforms.
- AzCopy is a command-line tool for copying blobs or files to or from a storage account.

Knowledge Check

Knowledge Check
01

Which of the following steps belongs to the cloud adoption framework?

A. Optimize

B. Migrate

C. Assess

D. All of these

Knowledge Check
01

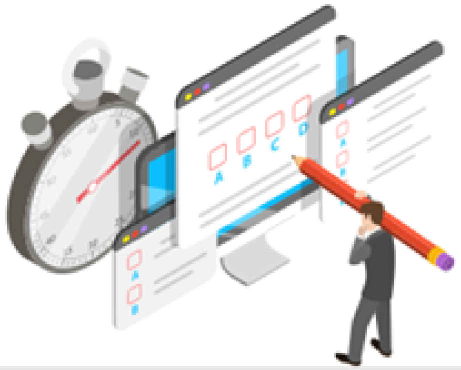
Which of the following steps belongs to the cloud adoption framework?

A. Optimize

B. Migrate

C. Assess

D. All of these



The correct answer is **D**

The cloud adoption framework consists of Assess, Migrate, Optimize, and Monitor.

Knowledge Check
02

Which command-line tool is used for copying data to and from Azure Blob storage, Azure Files, and Azure Table storage?

A. Sync

B. AzCopy

C. Copy

D. Transfer

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Knowledge Check

02

Which command-line tool is used for copying data to and from Azure Blob storage, Azure Files, and Azure Table storage?

A. Sync

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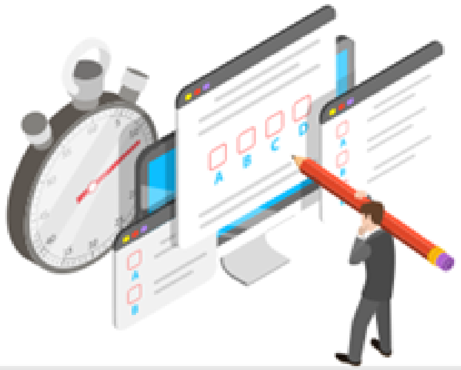


The correct answer is **B**

AzCopy is a command-line tool for copying data to or from Azure Blob storage, Azure Files, and Azure Table storage.

Knowledge Check
03

What are the benefits of Azure File Sync?

A. Cloud tiering

B. Multi-site access and sync

C. Business continuity and disaster recovery

D. All the above

Knowledge Check
03

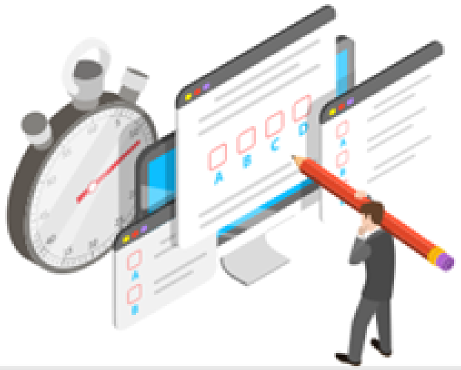
What are the benefits of Azure File Sync?

A. Cloud tiering

B. Multi-site access and sync

C. Business continuity and disaster recovery

D. All the above



The correct answer is **D**

Azure File Sync offers cloud tiering by arranging most frequent and least frequently accessed files, multi-site access, and business continuity by offering several redundancy options.

Knowledge Check
04

Which of the following migration options does Azure provide for an application?

A. Refactor

B. Rebuild

C. Rehost

D. All the above

Knowledge Check
04

Which of the following migration options does Azure provide for an application?

A. Refactor

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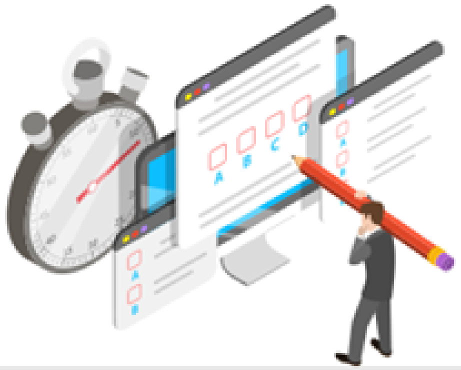


The correct answer is **D**

For each application, there are multiple migration options like rehost, refactor, rearchitect, rebuild, and replace.

Knowledge Check 05

Which of the following statements is correct about network security groups, route tables, and reserved IPs that are not attached to any virtual machines or virtual networks?

A. They cannot be migrated

B. They can be migrated independently

C. Both A and B

D. None of the above

Knowledge Check
05

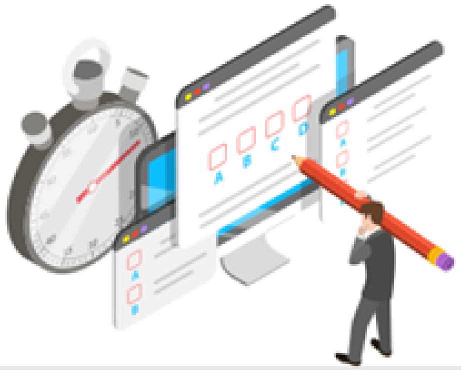
Which of the following statements is correct about network security groups, route tables, and reserved IPs that are not attached to any virtual machines or virtual networks?

A. They cannot be migrated

B. They can be migrated independently

C. Both A and B

D. None of the above



The correct answer is **B**

Network security groups, route tables, and reserved IPs that are not attached to any virtual machines and virtual networks can also be migrated independently.

TECHNOLOGY

Thank You